

Physics 829: Homework Set No. 1

**Due date: Friday, April 6, 2012, 1:00pm
in PRB M2043 (Biao Huang's office)**

Total point value of set: 60 points

Problem 1 (20 pts.): Obtain a variational estimate for the energy of a nonrelativistic particle moving in 1 dimension in the potential $V(x) = \lambda x^4$, using the trial wave function $\psi(x; \alpha) = e^{-\alpha x^4}$, and compare it with the estimate based on a Gaussian trial function (see (16.1.5) in Shankar). (*Hint:* Use a suitable variable transformation to convert the expectation value integrals into Gaussian integrals that you can do analytically. You should find that the two upper limits for the ground state energy agree with each other roughly within 10%. Which of the two is the better estimate you have to find out for yourself.)

Problem 2 (10 pts.): Use the variational method to estimate the energy of the first excited state for a nonrelativistic particle moving in 1 dimension in the potential $V(x) = \lambda x^4$. What are the expected general features of the wave function for this state? Make a suitable 1-parameter ansatz for the trial wave function, inspired by the ansatz (16.1.3) in Shankar for the ground state, but ensuring that the first excited state is always orthogonal on the ground state. Compute and minimize the energy expectation value in your trial wave function. Compare with Eqs. (16.1.4/5) and interpret your result.

Problem 3 (10 pts.): Show the work needed to arrive at the right hand side of Eq. (16.1.15) in Shankar.

Problem 4 (10 pts.): Exercise 16.1.5 (Shankar, p. 435)

Problem 5 (10 pts.): Exercise 16.2.4 (Shankar, p. 445)

Problem 1

$$V(x) = \lambda x^4 \quad \text{potential}$$

$$\psi(x; \alpha) = e^{-\alpha x^4} \quad \text{trial function}$$

$$E(\alpha) = \frac{\int_0^{\infty} dx e^{-\alpha x^4} \left(-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + \lambda x^4 \right) e^{-\alpha x^4}}{\int_0^{\infty} dx e^{-2\alpha x^4}}$$

Try to convert into Gaussian integrals:

$$u = x^2 \quad du = 2x dx \quad dx = \frac{du}{2x} = \frac{du}{2\sqrt{u}}$$

$$\frac{d}{dx} = \frac{du}{dx} \frac{d}{du} = 2x \frac{d}{du} = 2\sqrt{u} \frac{d}{du}$$

$$\frac{d^2}{dx^2} = 2\sqrt{u} \frac{d}{du} 2\sqrt{u} \frac{d}{du} = 4\sqrt{u} \left(\frac{1}{2\sqrt{u}} \frac{d}{du} + \sqrt{u} \frac{d^2}{du^2} \right) = 2 \frac{d}{du} + 4u \frac{d^2}{du^2}$$

$$\begin{aligned} \frac{d^2}{dx^2} e^{-\alpha x^4} &= \left(2 \frac{d}{du} + 4u \frac{d^2}{du^2} \right) e^{-\alpha u^2} = -4\alpha u e^{-\alpha u^2} + 4u \frac{d}{du} (-2\alpha u e^{-\alpha u^2}) \\ &= -4\alpha u e^{-\alpha u^2} - 8\alpha u e^{-\alpha u^2} + 4u 4\alpha^2 u^2 e^{-\alpha u^2} \\ &= (16\alpha^2 u^3 - 12\alpha u) e^{-\alpha u^2} \end{aligned}$$

$$E(\alpha) = \frac{\int_0^{\infty} \frac{du}{\sqrt{u}} e^{-\alpha u^2} \left(-\frac{\hbar^2}{2m} (16\alpha^2 u^3 - 12\alpha u) + \lambda u^2 \right) e^{-\alpha u^2}}{\int_0^{\infty} \frac{du}{\sqrt{u}} e^{-2\alpha u^2}}$$

$$\begin{aligned} (2\alpha u^2 = \xi^2) &= \frac{\int_0^{\infty} \frac{d\xi}{\sqrt{\xi}} e^{-\xi^2} \left(-\frac{\hbar^2}{2m} (4\sqrt{2\alpha} \xi^3 - 6\sqrt{2\alpha} \xi) + \frac{\lambda}{2\alpha} \xi^2 \right)}{\int_0^{\infty} \frac{d\xi}{\sqrt{\xi}} e^{-\xi^2}} \end{aligned}$$

$$= \frac{\int_0^{\infty} \frac{d\xi}{\sqrt{\xi}} e^{-\xi^2} \left(-\frac{\hbar^2 \sqrt{2\alpha}}{m} (2\xi^3 - 3\xi) + \frac{\lambda}{2\alpha} \xi^2 \right)}{\int_0^{\infty} \frac{d\xi}{\sqrt{\xi}} e^{-\xi^2}}$$

We see that the kinetic energy scales with $\sqrt{\alpha}$, the potential energy with $\frac{1}{\alpha}$.

We get

$$E(\alpha) = \left(-\frac{2\hbar^2\sqrt{2\alpha}}{m} I_{5/2} + \frac{3\hbar^2\sqrt{2\alpha}}{m} I_{1/2} + \frac{\lambda}{2\alpha} I_{3/2} \right) / I_{-1/2}$$

where

$$I_\nu \equiv \int_0^\infty d\xi \xi^\nu e^{-\xi^2} = \frac{1}{2} \Gamma\left(\frac{\nu+1}{2}\right)$$

$$\frac{I_{5/2}}{I_{-1/2}} = \frac{\Gamma(\frac{7}{4})}{\Gamma(\frac{1}{4})} = \frac{3}{4} \frac{\Gamma(\frac{3}{4})}{\Gamma(\frac{1}{4})} \equiv \frac{3}{4} \zeta$$

$$\frac{I_{3/2}}{I_{-1/2}} = \frac{\Gamma(\frac{5}{4})}{\Gamma(\frac{1}{4})} = \frac{1}{4}$$

$$\frac{I_{1/2}}{I_{-1/2}} = \frac{\Gamma(\frac{3}{4})}{\Gamma(\frac{1}{4})} \equiv \zeta = 0.337989$$

$$\Rightarrow E(\alpha) = \frac{\lambda}{8\alpha} + \frac{\hbar^2}{m} \sqrt{2\alpha} \zeta \left(3 - 2 \cdot \frac{3}{4} \right) = \frac{\lambda}{8\alpha} + \frac{3\hbar^2}{m} \sqrt{\frac{\alpha}{2}} \zeta$$

Minimize:

$$-\frac{\lambda}{8\alpha^2} + \frac{3\hbar^2}{m2\sqrt{2}\alpha} \zeta = 0 \Rightarrow \alpha^{3/2} = \frac{\lambda}{8} \frac{2\sqrt{2}m}{3\hbar^2 \zeta} = \frac{\lambda m}{\hbar^2} \frac{1}{6\sqrt{2}\zeta}$$

$$\alpha_0 = \left(\frac{\lambda m}{\hbar^2} \frac{1}{6\sqrt{2}\zeta} \right)^{2/3}$$

$$\left(\frac{m\lambda}{\hbar^2} \right)^{1/3} \text{ has units } (\text{length})^{-2}$$

$$\Rightarrow \frac{\lambda m}{\hbar^2} \sim (\text{length})^6 \Rightarrow \alpha_0 \text{ has units } (\text{length})^{-12/3} = (\text{length})^{-4} \checkmark$$

$$\begin{aligned}
E(\alpha_0) &= \frac{3\hbar^2}{\sqrt{2}m} \left(\frac{\lambda m}{\hbar^2} \frac{1}{6\sqrt{2}\eta} \right)^{1/3} + \frac{\lambda}{8} \left(\frac{\hbar^2}{\lambda m} 6\sqrt{2}\eta \right)^{2/3} \\
&= \left(\frac{27\hbar^6 \eta^3 \lambda m}{\sqrt{8} m^3 \hbar^2 6\sqrt{2}\eta} \right)^{1/3} + \left(\frac{\lambda^3 \hbar^4 36 \cdot 2 \cdot \eta^2}{2^9 \lambda^2 m^2} \right)^{1/3} \\
&= \left(\frac{9}{8} \frac{\hbar^4 \lambda}{m^2} \eta^2 \right)^{1/3} + \left(\frac{9}{2^6} \frac{\hbar^4 \lambda}{m^2} \eta^2 \right)^{1/3} = \left(\frac{9}{8} \frac{\hbar^4 \lambda}{m^2} \eta^2 \right)^{1/3} \left(1 + \frac{1}{2} \right) \\
&= \frac{3}{4} \left(\frac{6\hbar^4 \lambda}{m^2} \right)^{1/3} \left(\frac{3}{2} \eta^2 \right)^{1/3} \\
&= \frac{3}{8} \left(\frac{6\hbar^4 \lambda}{m^2} \right)^{1/3} (12\eta^2)^{1/3}
\end{aligned}$$

To decide whether this $e^{-\alpha x^4}$ trial function is a better or worse approximation to the ground state wavefunction than the Gaussian studied in class we have to compare the coefficients of $\left(\frac{6\hbar^4 \lambda}{m^2} \right)^{1/3}$; the answer boils down to whether $12\eta^2$ is smaller or larger than 1:

$$(12\eta^2)^{1/3} = 1.11$$

So the Gaussian gives a lower upper limit for E_0 and is thus a better trial function than the one explored in this problem.

From the comparison between the two trial functions we might hope that with the Gaussian trial function we may be within 10% of the true ground state energy, but there is no guarantee for this.

Problem 2

We want the lowest energy expectation value in the space orthogonal to the ground state.

We don't know the ground state exactly, so we take the Gaussian with the lowest energy as a substitute for the ground state, and minimize the energy in the space orthogonal to that Gaussian.

How should we parametrize the variational state whose minimal energy specimen will give us an approximation to the first excited state?

From general principles, the first excited state in an even potential should have negative parity and one node. To ensure normalizability and have an easy function to work with, let us give it a Gaussian factor that makes it drop to zero quickly at large $|x|$. Orthogonality to the Gaussian "pseudo-ground-state" is automatic for any odd wavefunction, since

$$\int_{-\infty}^{\infty} \psi_{\text{even}}(x) \psi_{\text{odd}}^*(x) dx = 0 \text{ always.}$$

So let's choose our variational state in the form

$$\tilde{\psi}_1(x; \alpha) = x e^{-\alpha x^2/2}$$

$$\Rightarrow \tilde{E}_1(\alpha) = \frac{\langle \tilde{\Psi}_1 | \hat{H} | \tilde{\Psi}_1 \rangle}{\langle \tilde{\Psi}_1 | \tilde{\Psi}_1 \rangle} = \frac{\int_{-\infty}^{\infty} dx x e^{-\alpha x^2/2} \left(-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + \lambda x^4 \right) x e^{-\alpha x^2/2}}{\int_{-\infty}^{\infty} dx x^2 e^{-\alpha x^2/2}}$$

$$= \frac{\int_0^{\infty} d\xi \xi e^{-\xi^2/2} \left(-\frac{\alpha \hbar^2}{2m} \frac{d^2}{d\xi^2} + \frac{\lambda}{\alpha^2} \xi^4 \right) \xi e^{-\xi^2/2}}{\int_0^{\infty} d\xi \xi^2 e^{-\xi^2/2}}$$

$$\frac{d}{d\xi} (\xi e^{-\xi^2/2}) = e^{-\xi^2/2} (1 - \xi^2)$$

$$\frac{d^2}{d\xi^2} (\xi e^{-\xi^2/2}) = (-2\xi) e^{-\xi^2/2} + (1 - \xi^2)(-\xi) e^{-\xi^2/2} = \xi e^{-\xi^2/2} (-2 - 1 + \xi^2)$$

$$= \xi (\xi^2 - 3) e^{-\xi^2/2}$$

$$\tilde{E}_1(\alpha) = \frac{\int_0^{\infty} d\xi \xi^2 e^{-\xi^2/2} \left(-\frac{\alpha \hbar^2}{2m} (\xi^2 - 3) + \frac{\lambda}{\alpha^2} \xi^4 \right)}{\int_0^{\infty} d\xi \xi^2 e^{-\xi^2/2}}$$

$$= \left[-\frac{\alpha \hbar^2}{2m} (I_4 - 3I_2) + \frac{\lambda}{\alpha^2} I_6 \right] / I_2$$

$$I_n = \int_0^{\infty} d\xi \xi^n e^{-\xi^2/2} = \frac{1}{2} \Gamma\left(\frac{n+1}{2}\right)$$

$$= \left[-\frac{\alpha \hbar^2}{2m} \left(\underbrace{\frac{\Gamma(\frac{5}{2})}{\Gamma(\frac{3}{2})}}_{3/2} - 3 \right) + \frac{\lambda}{\alpha^2} \underbrace{\frac{\Gamma(\frac{7}{2})}{\Gamma(\frac{5}{2})}}_{\frac{5 \cdot 3}{2 \cdot 2} = \frac{15}{4}} \right]$$

$$= \frac{3}{4} \frac{\alpha \hbar^2}{m} + \frac{15}{4} \frac{\lambda}{\alpha^2}$$

$$\frac{d\tilde{E}_1(\alpha)}{d\alpha} = 0 = \frac{3}{4} \frac{\hbar^2}{m} - \frac{15}{2} \frac{\lambda}{\alpha^3} \Rightarrow \alpha_1 = \frac{15\lambda}{2} \cdot \frac{4}{3} \frac{m}{\hbar^2}$$

$$= 10 \frac{\lambda m}{\hbar^2}$$

$$\boxed{\alpha_1 = \left(10 \frac{\lambda m}{\hbar^2} \right)^{1/3}}$$

The variational estimate for the energy of the first excited state is thus

$$\tilde{E}_1 \equiv \tilde{E}_1(\alpha_1) = \frac{3\hbar^2}{4m} \left(10 \frac{\lambda m}{\hbar^2}\right)^{1/3} + \frac{15}{4} \lambda \left(\frac{\hbar^4}{10^2 \lambda^2 m^2}\right)^{1/3}$$

$$= \frac{3}{4} \left[\left(10 \frac{\hbar^6 \lambda m}{\hbar^2 m^3}\right)^{1/3} + \left(\frac{.5^3 \cdot \lambda^3 \hbar^4}{\lambda^2 m^2 \cdot 5^2 \cdot 2^2}\right)^{1/3} \right]$$

$$= \frac{3}{4} \left(\frac{10 \hbar^4 \lambda}{m^2}\right)^{1/3} \left(1 + \frac{1}{2}\right) \approx \frac{9}{8} \left(\frac{10 \hbar^4 \lambda}{m^2}\right)^{1/3}$$

$$\Rightarrow \boxed{\tilde{E}_1 = \frac{9}{8} \left(\frac{10 \hbar^4 \lambda}{m^2}\right)^{1/3}}$$

Sanity check:

The ratio of the energy of the first excited state to that of the ground state, ($\tilde{E}_0 = \frac{3}{8} \left(\frac{6 \hbar^4 \lambda}{m^2}\right)^{1/3}$) is accordingly estimated as

$$\frac{\tilde{E}_1}{\tilde{E}_0} = 3 \left(\frac{10}{6}\right)^{1/3} \approx 3.56$$

This is a bit larger than the corresponding ratio for the harmonic oscillator potential which is 3.

Problem 3

Starting point is l.h.s. of (16.1.15):

$$E(Z) = \frac{\int d^3r_1 d^3r_2 \psi(r_1, r_2; Z) \left[-\frac{\hbar^2}{2m} (\nabla_1^2 + \nabla_2^2) - \frac{Ze^2}{r_1} - \frac{Ze^2}{r_2} + \frac{e^2}{|\vec{r}_1 - \vec{r}_2|} \right] \psi(r_1, r_2; Z)}{\int d^3r_1 d^3r_2 |\psi(r_1, r_2; Z)|^2}$$

with

$$\psi(r_1, r_2; Z) = \frac{Z^3}{\pi a_0^3} e^{-Z(r_1 + r_2)/a_0}$$

Let's split the energy into

$$E(Z) = T_1 + T_2 + V_1 + V_2 + V_{ee} \equiv E_1 + E_2 + V_{ee} \quad (E_i = T_i + V_i, i=1,2)$$

ψ is a product of eigenfunctions for the lowest eigenstates of a hydrogen-like atom with nuclear charge Z . Using the virial theorem for the Coulomb potential, $\langle \hat{T} \rangle_Z = -\frac{1}{2} \langle \hat{V} \rangle_Z$ for these states, we get

$$\underbrace{-\frac{m(Ze^2)^2}{2\hbar^2}}_{E_0(Z)} = \langle \hat{T} \rangle_Z + \langle \hat{V} \rangle_Z = \left\langle \frac{\hat{p}^2}{2m} - \frac{Ze^2}{\hat{r}} \right\rangle_Z = -\langle \hat{T} \rangle_Z$$

$$\Rightarrow \boxed{T_1 = T_2 = Z^2 R_y}$$

$$\begin{aligned} \text{Next, we also set } \boxed{V_1 = V_2} &= \left\langle -\frac{Ze^2}{r_i} \right\rangle_Z = \frac{2}{Z} \left\langle -\frac{Ze^2}{r_i} \right\rangle_Z \\ &= \frac{2}{Z} \langle \hat{V} \rangle_Z = \frac{2}{Z} (-2 \langle \hat{T} \rangle_Z) \\ &= \frac{2}{Z} (-2 Z^2 R_y) = \boxed{-4Z R_y} \end{aligned}$$

$$\Rightarrow \underline{E_1 + E_2} = \underline{T_1 + T_2 + V_1 + V_2} = \underline{2 \cdot (Z^2 - 4Z) R_y}$$

This gives the first two terms of the r.h.s. of (16.1.15)

The last term must therefore arise from the electron-electron interaction. Let's compute it:

$$V_{ee} = \frac{\int_0^\infty r_1^2 dr_1 \int_0^{2\pi} d\varphi_1 \int_{-1}^1 d\cos\vartheta_1 \int_0^\infty r_2^2 dr_2 \int_0^{2\pi} d\varphi_2 \int_{-1}^1 d\cos\vartheta_2 \cdot \frac{e^2}{|\vec{r}_1 - \vec{r}_2|} \cdot e^{-2Z(r_1+r_2)/a_0}}{\int_0^\infty r_1^2 dr_1 \int_0^{2\pi} d\varphi_1 \int_{-1}^1 d\cos\vartheta_1 \int_0^\infty r_2^2 dr_2 \int_0^{2\pi} d\varphi_2 \int_{-1}^1 d\cos\vartheta_2 e^{-2Z(r_1+r_2)/a_0}}$$

We write

$$\frac{1}{|\vec{r}_1 - \vec{r}_2|} = \frac{1}{\sqrt{r_1^2 + r_2^2 - 2r_1 r_2 \cos\vartheta_{12}}}$$

and orient our integration coord. system for $\vec{r}_2$ such that z_2 points in $\vec{r}_1$ direction $\Rightarrow \vartheta_{12} = \vartheta_2$

$$\Rightarrow \frac{1}{|\vec{r}_1 - \vec{r}_2|} = \frac{1}{\sqrt{r_1^2 + r_2^2 - 2r_1 r_2 \cos\vartheta_2}}$$

The integrals over φ_1, φ_2 and ϑ_1 are now trivial and yield a factor $8\pi^2$ that cancels between

numerator and denominator:

$$V_{ee} = \frac{\int_0^\infty r_1^2 dr_1 e^{-2Zr_1/a_0} \int_0^\infty r_2^2 dr_2 e^{-2Zr_2/a_0} \int_{-1}^1 dz \frac{r^2}{\sqrt{r_1^2 + r_2^2 - 2r_1 r_2 z}}}{\int_0^\infty r_1^2 dr_1 e^{-2Zr_1/a_0} \int_0^\infty r_2^2 dr_2 e^{-2Zr_2/a_0} \cdot 2}$$

$$\int_{-1}^1 \frac{dz}{\sqrt{r_1^2 + r_2^2 - 2r_1 r_2 z}} = \left. \frac{2\sqrt{r_1^2 + r_2^2 - 2r_1 r_2 z}}{-2r_1 r_2} \right|_{-1}^1 = -\frac{|\vec{r}_1 - \vec{r}_2| - (r_1 + r_2)}{r_1 r_2} = \frac{r_1 + r_2 - |\vec{r}_1 - \vec{r}_2|}{r_1 r_2}$$

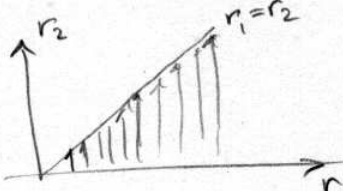
$$= \begin{cases} \frac{2}{r_1} & r_1 > r_2 \\ \frac{2}{r_2} & r_2 > r_1 \end{cases}$$

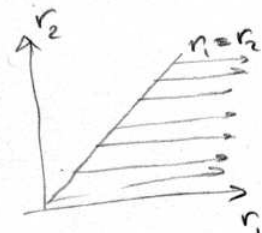
$$\Rightarrow \int_{-1}^1 dz \frac{e^2}{\sqrt{r_1^2 + r_2^2 - 2r_1 r_2 \cos \theta}} = \frac{2e^2}{r_>}$$

where $r_>$ is the larger of r_1 and r_2

$$\Rightarrow V_{ee} = \frac{\int_0^\infty r_1^2 dr_1 \int_0^\infty r_2^2 dr_2 \frac{e^2}{r_>} e^{-2Z(r_1+r_2)/a_0}}{\int_0^\infty r_1^2 dr_1 \int_0^\infty r_2^2 dr_2 e^{-2Z(r_1+r_2)/a_0}}$$

$$= e^2 \frac{\int_0^\infty r_1^2 dr_1 \int_{r_1}^\infty r_2 dr_2 e^{-2Z(r_1+r_2)/a_0} + \int_0^\infty r_1 dr_1 \int_0^{r_1} r_2^2 dr_2 e^{-2Z(r_1+r_2)/a_0}}{\int_0^\infty r_1^2 dr_1 \int_0^\infty r_2^2 dr_2 e^{-2Z(r_1+r_2)/a_0}}$$

$$\int_0^\infty r_1 dr_1 \int_0^{r_1} dr_2 = \int_0^\infty r_2 dr_2 \int_{r_2}^\infty dr_1$$


$$= \int_0^\infty dr_2 \int_{r_2}^\infty dr_1$$


$$\Rightarrow \int_0^\infty r_1 dr_1 e^{-2Zr_1/a_0} \int_0^{r_1} r_2^2 dr_2 e^{-2Zr_2/a_0} = \int_0^\infty dr_2 r_2^2 e^{-2Zr_2/a_0} \int_{r_2}^\infty r_1 dr_1 e^{-2Zr_1/a_0}$$

This makes the second integral equal to the first one in V_{ee} .

$$\text{Use } \xi = \frac{2Zr}{a_0} \Rightarrow r = \frac{a_0 \xi}{2Z}$$

$$\Rightarrow V_{ee} = \frac{2Ze^2}{a_0} \cdot 2 \cdot \frac{\int_0^\infty d\xi_1 \xi_1^2 e^{-\xi_1} \int_{\xi_1}^\infty d\xi_2 \xi_2 e^{-\xi_2}}{(\int_0^\infty d\xi \xi^2 e^{-\xi})^2}$$

$$\int_{\xi_1}^\infty d\xi_2 \xi_2 e^{-\xi_2} = -e^{-\xi_1} (\xi_1 + 1) \Big|_{\xi_1}^\infty = e^{-\xi_1} (\xi_1 + 1)$$

$$\int_0^\infty d\xi_1 (\xi_1^3 + \xi_1^2) e^{-2\xi_1} = \underbrace{-\frac{1}{2} \xi_1^3 e^{-2\xi_1}}_0 \Big|_0^\infty + \frac{5}{2} \int_0^\infty d\xi_1 \xi_1^2 e^{-2\xi_1}$$

$$= \frac{5}{2} e^{-2\xi} \left(-\frac{\xi^2}{2} - \frac{2\xi}{4} - \frac{2}{8} \right) \Big|_0^\infty = \frac{5}{8}$$

$$\int_0^{\infty} d\zeta \zeta^2 e^{-\zeta} = e^{-\zeta} (-\zeta^2 - 2\zeta - 2) \Big|_0^{\infty} = 2$$

$$\left(\int_0^{\infty} d\zeta \zeta^n e^{-a\zeta} = \frac{\Gamma(n+1)}{a^{n+1}} \right)$$

$$\Rightarrow V_{ee} = \frac{4Ze^2}{a_0} \cdot \frac{5}{8} \cdot \frac{1}{4} = \frac{Ze^2 me^2}{2\hbar^2} \frac{5}{4} = \frac{5}{4} Z R_y$$

Hence the total energy in the state $\psi(n_1, n_2; Z)$ is

$$\underline{E(Z) = 2R_y \left(Z^2 - 4Z + \frac{5}{8}Z \right) = -2R_y \left(4Z - Z^2 - \frac{5}{8}Z \right)}$$

as desired.

Exercise 16.1.5 (p. 435)

$$\hat{H} = \frac{\hat{p}^2}{2\mu} - \frac{e^2}{r}, \quad l=1$$

Trial function should behave as $R_{ne}(r) \sim r^l = r$ (for $l=1$) and have the smallest possible number of nodes (namely zero) to minimize kinetic energy.

At $r \rightarrow \infty$, the correct asymptotic behavior for R_{ne} in the Coulomb potential is $R_{ne}(r) \sim e^{-\kappa r}$ with $\kappa = \sqrt{2\mu|E|/\hbar^2}$.

We don't know E , so we treat κ as a variational parameter.

So we use as trial function

$$\psi(r, \vartheta, \varphi; \kappa, m) = r e^{-\kappa r} Y_{1m}(\vartheta, \varphi)$$

and minimize

$$E(\kappa, m) = \frac{\int_0^\infty r^2 dr \int d\Omega \, r e^{-\kappa r} Y_{1m}^*(\vartheta, \varphi) \left(-\frac{\hbar^2}{2\mu} \nabla^2 - \frac{e^2}{r} \right) r e^{-\kappa r} Y_{1m}(\vartheta, \varphi)}{\int d\Omega \int_0^\infty r^2 dr \, r^2 e^{-2\kappa r} |Y_{1m}(\vartheta, \varphi)|^2}$$

In the denominator, $\int d\Omega |Y_{1m}(\vartheta, \varphi)|^2 = 1$.

In the numerator, $\nabla^2 = \frac{d^2}{dr^2} + \frac{2}{r} \frac{d}{dr} - \frac{1}{r^2} \hat{L}^2$

When $\hat{L}^2$ acts on Y_{lm} it gives eigenvalue $l(l+1)\hbar^2$, independent of m . So all m values give the same

energy, $E(\kappa, m) = E(\kappa)$ independent of m , and the

variational energy eigenvalue will be 3-fold degenerate ($m = \pm 1, 0$).

$$\Rightarrow E(k) = \frac{\int_0^\infty r^2 dr \left(\int d\Omega |Y_{lm}(\theta, \varphi)|^2 \right) r e^{-kr} \left(-\frac{\hbar^2}{2\mu} \left(\frac{d^2}{dr^2} + \frac{2}{r} \frac{d}{dr} - \frac{2}{r^2} \right) - \frac{e^2}{r} \right) r e^{-kr}}{\int_0^\infty r^2 dr r^2 e^{-2kr}}$$

$$\frac{d}{dr} (r e^{-kr}) = (1 - kr) e^{-kr}$$

$$\frac{d^2}{dr^2} (r e^{-kr}) = -k e^{-kr} + (1 - kr)(-k) e^{-kr} = -k(2 - kr) e^{-kr}$$

$$\begin{aligned} \left(\frac{d^2}{dr^2} + \frac{2}{r} \frac{d}{dr} - \frac{2}{r^2} \right) (r e^{-kr}) &= \left(-k(2 - kr) + \frac{2}{r} (1 - kr) - \frac{2}{r} \right) e^{-kr} \\ &= (-2k + k^2 r - 2k) e^{-kr} = \frac{1}{r} e^{-kr} ((kr)^2 - 4kr) \end{aligned}$$

$$\Rightarrow E(k) = \frac{\int_0^\infty r^2 dr e^{-2kr} \left(-\frac{\hbar^2}{2\mu} ((kr)^2 - 4kr) - r e^2 \right)}{\int_0^\infty r^2 dr r^2 e^{-2kr}}$$

$$(\xi = 2kr) = \frac{\int_0^\infty \xi^2 d\xi e^{-\xi} \left(-\frac{\hbar^2}{2\mu} \left(\frac{\xi^2}{4} - 2\xi \right) - \frac{e^2 \xi}{2k} \right)}{\int_0^\infty \xi^2 d\xi e^{-\xi} \frac{\xi^2}{4k^2}} \quad \left(\int_0^\infty \xi^n e^{-\xi} d\xi = n! \right)$$

$$= \frac{4k^2}{4!} \left(-\frac{\hbar^2}{2\mu} \left(\frac{4!}{4} - 2 \cdot 3! \right) - \frac{e^2}{2k} 3! \right)$$

$$= + \frac{\hbar^2}{2\mu} k^2 - \frac{e^2 k^2}{2k} =$$

$$\left(\frac{\hbar^2}{\mu} = e^2 a_0 \right)$$

$$= \frac{1}{2} e^2 a_0 k^2 - \frac{1}{2} e^2 k$$

$$\boxed{E(k) = \frac{e^2}{2} (a_0 k^2 - k)}$$

$$\text{Minimize: } \frac{dE}{dk} = \frac{e^2}{2} (2a_0 k - 1) = 0 \Rightarrow$$

$$\boxed{k_0 = \frac{1}{2a_0}}$$

$$\Rightarrow \boxed{\psi(r, \vartheta, \varphi; l=1, k_0, m) = r e^{-r/2a_0} Y_{1m}(\vartheta, \varphi)}$$

Optimized wave function (not normalized)

The variational energy bound becomes

$$E_{l=1}(k_0) = \frac{e^2}{2} \left(a \frac{1}{4a_0^2} - \frac{1}{2a_0} \right) = -\frac{e^2}{8a_0} = -\frac{R_y}{4} = E_{n=2}$$

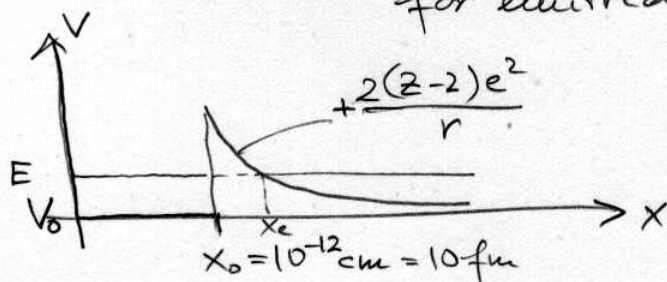
Due to careful choice of our trial wavefunction, taking into account everything we know about the asymptotic behavior of the wave function at $r \rightarrow 0$ and $r \rightarrow \infty$ and minimizing the number of nodes, the variational method has led us to the exact wavefunction and binding energy of the lowest $l=1$ state in hydrogen. It also gave us the correct degeneracy, $2l+1=3$, for this state.

Exercise 16.2.4. (p. 445)

$T_\alpha = 4.2 \text{ MeV}$ (kinetic energy of α particle emitted in α -decay of ^{92}U)

"Assume that $V(x)$ is just Coulombic for $x \geq x_0$ "

$\Rightarrow$ orbital angular momentum $l=0$ for emitted α -particle



$$\begin{aligned} V(x) &= \frac{(Z_1 e)(Z_2 e)}{x} \\ &= \frac{2(Z_u - 2)e^2}{x} \\ &= \frac{180e^2}{x} \quad \text{for } x \geq x_0 \end{aligned}$$

At $x \rightarrow 0$ all energy of the α -particles is kinetic ($V(x) \xrightarrow{x \rightarrow \infty} 0$) $\Rightarrow E = 4.2 \text{ MeV}$

The lifetime is $\tau = \frac{1}{R} = e^\gamma \frac{2m x_0}{\sqrt{2m E}} = e^\gamma \cdot 2 \times 4.003 \times 931.5 \frac{\text{MeV}}{c^2}$

$$= e^\gamma \cdot \frac{2 \times 4.003 \times 931.5 \frac{\text{MeV}}{c^2} \times 10^{-14} \text{ m}}{\sqrt{2 \times 4.003 \times 931.5 \frac{\text{MeV}}{c^2} \cdot 4.2 \text{ MeV}}}$$

$$= e^\gamma \cdot \frac{2 \times 4.003 \times 931.5 \times 10}{\sqrt{2 \times 4.003 \times 931.5 \times 4.2}} \frac{\text{fm}}{c} = e^\gamma \times 3 \times 10^{-23} \text{ s} \times \underbrace{\sqrt{\frac{2 \times 4 \times 931.5}{4.2}}}_{42.1}$$

$$= e^\gamma \times 1.26 \times 10^{-21} \text{ s}$$

$$\left(\frac{1 \text{ fm}}{c} = \frac{10^{-15} \text{ m}}{3 \times 10^8 \text{ m/s}} = 3 \times 10^{-24} \text{ s} \right)$$

The tunneling factor is

$$\gamma = \frac{2}{\hbar} \int_{x_0}^{x_e} dx \sqrt{2m (V(x) - E)} = \frac{2\sqrt{2m}}{\hbar} \int_{x_0}^{x_e} dx \sqrt{\frac{180e^2}{x} - E}$$

The classical turning point x_e is obtained from

$$\frac{180e^2}{x_e} = E \Rightarrow x_e = \frac{2Ze^2}{E} \text{ with } Z=90$$

$$\Rightarrow \gamma = \frac{2\sqrt{2mE}}{\hbar} \int_{x_0}^{x_e} dx \sqrt{\frac{x_e}{x} - 1} \quad \gamma = \frac{x}{x_e}$$

$$= \frac{2\sqrt{2mE}x_e}{\hbar} \int_{x_0/x_e}^1 dz \sqrt{\frac{1}{z} - 1}$$

$$= \frac{4Ze^2}{\hbar} \frac{1}{\sqrt{E/2m}} \int_y^1 dz \sqrt{\frac{1}{z} - 1}$$

$$y \equiv \frac{x_0}{x_e} < 1$$

$$\frac{E^2}{2m} = \frac{p^2}{(2m)^2} = \frac{1}{4} \left(\frac{p}{m} \right)^2$$

$$= \frac{8Ze^2}{\hbar v} \int_y^1 dz \sqrt{\frac{1}{z} - 1}$$

Let us substitute $z = \cos^2 t \Rightarrow dz = -2 \cos t \sin t dt$

$$\sqrt{\frac{1}{z} - 1} = \sqrt{\frac{z-1}{z}} = \sqrt{\frac{\cos^2 t - 1}{\cos^2 t}} = \frac{\sin t}{\cos t}$$

$$\int_y^1 dz \sqrt{\frac{1}{z} - 1} = -2 \int_{\cos^{-1}\sqrt{y}}^0 dt \sin t \cos t \frac{\sin t}{\cos t}$$

$$= 2 \int_0^{\cos^{-1}\sqrt{y}} dt \sin^2 t = 2 \left[\frac{t}{2} - \frac{1}{4} \sin 2t \right]_0^{\cos^{-1}\sqrt{y}}$$

$$= \cos^{-1}\sqrt{y} - \sin t \cos t \Big|_0^{\cos^{-1}\sqrt{y}} = \cos^{-1}\sqrt{y} - \cos t \sqrt{1 - \cos^2 t} \Big|_0^{\cos^{-1}\sqrt{y}}$$

$$= \cos^{-1}\sqrt{y} - \sqrt{y}\sqrt{1-y} \text{ as desired}$$

$$\text{Now } \gamma = \frac{x_0}{x_e} = \frac{10 \text{ fm} \times 4.2 \text{ MeV}}{180 e^2} = \frac{42 \text{ MeV fm} \times 137}{180 \times 197.33 \text{ MeV fm}} \quad \boxed{e^2 = \frac{1}{137} \hbar c = \frac{197.33 \text{ MeV fm}}{137}}$$

$$= 0.16 \approx \frac{1}{6} \ll 1$$

For small $\sqrt{\gamma}$, $\cos^{-1} \sqrt{\gamma} = \frac{\pi}{2} - \sqrt{\gamma} - \frac{(\sqrt{\gamma})^3}{2 \cdot 3} - \frac{3(\sqrt{\gamma})^5}{2 \cdot 4 \cdot 5} - \dots$

$$= 1.5708 - 0.4025 - 0.0109 - 0.0008 = 1.157$$

$$\Rightarrow \cos^{-1} \sqrt{\gamma} - \sqrt{\gamma} \sqrt{1-\gamma} = 0.79$$

$$\Rightarrow \gamma = 0.79 \frac{4Ze^2}{\hbar c} \sqrt{\frac{2mc^2}{E}} = 0.79 \times 4 \times \frac{90}{137} \underbrace{\sqrt{\frac{2 \times 4.003 \times 931.5}{4.2}}}_{42.1} = 87.5$$

Plugging this into the lifetime:

$$\tau = 1.26 \times 10^{-21} \times e^{0.4343 \times 87.5 \log e} \text{ s} = 1.26 \times 10^{-21} \times 10^{0.4343 \times 87.5}$$

$$= 1.26 \times 10^{-21} \times 10^{38} \text{ s} = 1.26 \times 10^{17} \text{ s} \cdot \frac{1 \text{ yr}}{3.16 \times 10^7 \text{ s}} = 4 \times 10^9 \text{ years}$$

$$\boxed{\tau = 4 \times 10^9 \text{ yr}}$$

The lifetime of a $^{238}_{92}\text{U}$ against α -decay with 4.2 MeV α -particles is about 4 billion years.

(According to Wikipedia, $^{238}_{92}\text{U}$ decays in ^{234}Th by emission of 4.27 MeV α -particles with a lifetime of $4.468 \times 10^9 \text{ yr}$ - we are pretty close!)